5. Find formulas for the highest exponent e of the prime p such that p^e divides (*a*) the product $2 \cdot 4 \cdot 6 \cdots (2n)$ of the first n even numbers; (*b*) the product of the first n odd numbers.

$$p^e \parallel 2^n \cdot (n!) \Rightarrow \begin{cases} p^{e_1} \parallel 2^n \Rightarrow e_1 = \begin{cases} n & (p=2) \\ 0 & (p>2) \end{cases} \\ p^{e_2} \parallel (n!) \Rightarrow e_2 = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor \end{cases}$$

$$\Rightarrow e = \begin{cases} n + \sum_{k=1}^{\infty} \left\lfloor \frac{n}{2^k} \right\rfloor & (p=2) \\ \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor & (p>2) \end{cases}$$

$$p^{e+f} \parallel (2n)! \Rightarrow p^f \parallel \frac{(2n)!}{2^n \cdot (n!)} \Rightarrow e+f = \sum_{k=1}^{\infty} \left\lfloor \frac{2n}{p^k} \right\rfloor$$

$$\Rightarrow f = \begin{cases} \sum_{k=1}^{\infty} \left\lfloor \frac{2n}{2^k} \right\rfloor - n - \sum_{k=1}^{\infty} \left\lfloor \frac{n}{2^k} \right\rfloor = 0 & (p=2) \\ \sum_{k=1}^{\infty} \left\lfloor \frac{2n}{p^k} \right\rfloor - \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor & (p>2) \end{cases}$$

***9.** Prove that $(2n)!/(n!)^2$ is even if n is a positive integer.

2021

-10585

Note. $\frac{(2n)!}{(n!)(n!)} = \binom{2n}{n}$

분자에 2의 개수 : $\sum_{k=1}^{\infty} \left\lfloor \frac{2n}{2^k} \right\rfloor = n + \sum_{k=1}^{\infty} \left\lfloor \frac{n}{2^k} \right\rfloor$

분모에 2의 개수 : $2 \cdot \sum_{k=1}^{\infty} \left\lfloor \frac{n}{2^k} \right\rfloor$

수학과 2학년 이상

claim $\sum_{k=1}^{\infty} \left\lfloor \frac{n}{2^k} \right\rfloor < \sum_{k=1}^{\infty} \frac{n}{2^k} = n$

equality requires $2^k \mid n$ for $\forall k \in \mathbb{N}$.

but since n is finite natural number, it is impossible.

***13.** If a and b are positive integers such that $(a, b) = 1$, and ρ is a real number such that $a\rho$ and $b\rho$ are integers, prove that ρ is an integer. Hence prove that $\rho = n!/(a!b!)$ is an integer if $(a, b) = 1$ and $a + b = n + 1$. Generalize this to prove that

$$\rho = \frac{n!}{a_1!a_2!\cdots a_r!}$$

is an integer if $(a_1, a_2, \cdots, a_r) = 1$ and $a_1 + a_2 + \cdots + a_r = n + 1$. [Note that the first part of this problem implies that the binomial coefficient $\binom{m}{a}$ is divisible by m if $(m, a) = 1$. This follows by writing $n = m - 1$, so that $(a, b) = 1$ is equivalent to $(a, m) = 1$.]

suppose $a\rho = x$, $b\rho = y$

(x,y)로 나눈다.

$\rho = \frac{x}{a} = \frac{y}{b}$ with $a, b \in \mathbb{N}$, $x, y \in \mathbb{Z}$, $(a,b) = 1$

$\rho' = \frac{x'}{a} = \frac{y'}{b}$

$ay' = bx'$ implies $a \mid x'$, $b \mid y'$

therefore ρ' and ρ are integers

$a\rho = a \cdot \frac{n!}{a!\,b!} = \frac{n!}{(a-1)!\,b!} = \binom{n}{b} \in \mathbb{Z}$

$b\rho = b \cdot \frac{n!}{a!\,b!} = \frac{n!}{a!\,(b-1)!} = \binom{n}{a} \in \mathbb{Z}$

with $(a,b) = 1$, ρ is integer.

$a_1 + a_2 + \cdots + a_r = n+1$

$a_1x_1 + a_2x_2 + \cdots + a_rx_r = 1$

i.e $(a_1, \ldots, a_r) = 1$

relatively prime

$(a_1\rho)x_1 + (a_2\rho)x_2 + \cdots + (a_r\rho)x_r = \rho$

with $a_k\rho = \frac{n!}{a_1! \cdots (a_k - 1)! \cdots a_r!} \in \mathbb{Z}$

Multinomial Coefficient.

since $(a_k\rho)$ and x_k are both integers

ρ is also integers

7. Prove that $\sigma_{-k}(n) = n^{-k}\sigma_k(n)$.

$\sigma_k(n) = \sum_{d|n} d^k$ Claim $\sigma_{-k}(n) = n^{-k}\sigma_k(n)$

$\because$ $n^{-k}\sum_{d|n} d^k = \sum_{d|n} (n/d)^{-k}$

$= \sum_{d'|n} d'^{-k}$ $(d' = n/d)$

d'는 n의 모든 divisor들을 한번씩 훑는다.

---

8. Find a formula for $\sigma_k(n)$.

- $n = p_1^{j_1} \cdots p_l^{j_l}$
- $\sum_{d|n} d^k = (1 + p_1^k + \cdots p_1^{j_1 k}) \cdots (1 + p_l^k + \cdots + p_l^{j_l k})$

$= \prod_{p^j \| n} (1 + p^k + \cdots + p^{jk}) = \prod_{p^j \| n} \frac{(p^k)^{j+1} - 1}{p^k - 1}$

---

10. Give an example to show that if $f(n)$ is totally multiplicative, $F(n)$ need not also be totally multiplicative, where $F(n)$ is defined as $\sum_{d|n} f(d)$.

let $(n, m) = 1$ then with $f(n) = 1$ $F(16) \neq (F(4))^2$

$F(n) = \sum_{d|n} 1$ $F(4) = 3$ $\because$ $1|4$ $2|4$ $4|4$

$F(16) = 5$ $\because$ $1|16$ $2|16$ $4|16$ $8|16$ $16|16$

---

16. We say (following Euclid) that m is a *perfect number* if $\sigma(m) = 2m$, that is, if m is the sum of all its positive divisors other than itself. If $2^n - 1$ is a prime p, prove that $2^{n-1}p$ is a perfect number. Use this result to find three perfect numbers.

$\sigma(2^{n-1}p) = (1 + 2 + \cdots + 2^{n-1})(1 + p)$

$= \underbrace{(2^n - 1)}_{=p}(2^n) = 2 \cdot (2^{n-1} \cdot p)$

perfect.

$p_1 = 2^2 - 1 = 3 \rightsquigarrow 2^1 \cdot 3 = 6$

$p_2 = 2^3 - 1 = 7 \rightsquigarrow 2^2 \cdot 7 = 28$

$p_3 = 2^5 - 1 = 31 \rightsquigarrow 2^4 \cdot 31 = 496$

3. Evaluate $\sum_{j=1}^{\infty} \mu(j!)$.

- $\mu(1) = +1$ $\quad n \geq 4$ then $2^2 \mid n!$

  $\mu(2!) = -1$ $\quad \mu(n!) = 0$

  $\mu(3!) = +1$

- therefore $\sum_{j=1}^{\infty} \mu(j!) = +1$

***26.** Let $\Phi_n(x)$ denote the polynomial with leading coefficient 1 and degree $\phi(n)$ whose roots are the $\phi(n)$ different primitive nth roots of unity. Prove that $\prod_{d|n} \Phi_d(x) = x^n - 1$ for all real or complex numbers x. Deduce that $\Phi_n(x) = \prod_{d|n}(x^d - 1)^{\mu(n/d)}$. Show that the coefficients of $\Phi_n(x)$ are integers. This is the *cyclotomic polynomial* of order n.

- $x^n = 1$ means $\quad$ then $x^n - 1 = \prod_{d|n} \Phi_d(x)$

  $\exists d \in \mathbb{N}$ s.t. $\begin{cases} d \mid n \\ x^d = 1 \end{cases}$ $\quad$ i.e. order가 d인 근은

  오직 $\Phi_d(x)$ 의 근이 된다.

- 오일러 함수는 관련 공식 $\quad \Phi_n(x) = \sum_{d|n} (x^d - 1)^{\mu(n/d)}$

  나눗셈.

  ↓

- $\Phi_1(x) = (x-1)$ $\Longrightarrow$ $\underbrace{x^n - 1}_{\text{정수 계수}} = \Phi_n(x) \underbrace{\left(\prod_{\substack{d|n \\ d<n}} \Phi_d(x) \right)}_{\text{오일러 함수}}$

  1~ 정수 계수 (induction)

Theorem 4.9 *If $f(n) = \sum_{d|n} \mu(d) F(n/d)$ for every positive integer n, then $F(n) = \sum_{d|n} f(d)$.*

$$F(n) = \sum_{d'|n} F(d') \, \mu(n/d')$$

$n = d'$

$$= \sum_{d|n} \sum_{d'|n/d} \mu(n/dd') F(d')$$

$$= \sum_{d|n} f(n/d)$$

***29.** Let p be prime, and let $\Phi_{p-1}(x)$ denote the cyclotomic polynomial of order $p-1$. Show that g is a solution of the congruence $\Phi_{p-1}(x) \equiv 0 \pmod{p}$ if and only if g is a primitive root $\pmod{p}$. Slow also that the sum of all the primitive roots $\pmod{p}$ is $\equiv \mu(p-1) \pmod{p}$.

- (Fermat) primitive root g (p)

$$\Rightarrow \quad 0 \equiv x^{p-1} - 1 \pmod{p}$$

$$= \prod_{d \mid p-1} \Phi_d(x)$$

- $f(x) = x^{p-1} - 1$

$f'(x) = (p-1)x^{p-2} \not\equiv 0$ if $x \not\equiv 0 \pmod{p}$

therefore $f(x)$ $f'(x)$는 공통근 無 $\pmod{p}$

$\Rightarrow$ $(x^{p-1}-1) \equiv 0$ 은 정수하 $p-1$개의 다른 근을 가짐.

$\Rightarrow$ $\Phi_d(x) \pmod{p}$ 은 근을 공유하지 않음.

$\Leftarrow$

g가 primitive root이면

$\Phi_{p-1}(x)$의 해가 된다.

- $a \not\equiv 0$의 order k $\pmod{p}$

(i) $k \mid p-1$

(ii) a는 $x^k - 1 \equiv 0$ (p)의 근

(iii) a는 $\Phi_k(x)$의 근 (공통근 없음)

$\Rightarrow$

g가 $\Phi_{p-1}(x)$의 해가려면

위수 $k \mid p-1$

(i) $1 < k < p-1$ 라면 g는 Φ_k의 근.

$\rightarrow\leftarrow$

(ii) $k = p-1$

***29.** Let p be prime, and let $\Phi_{p-1}(x)$ denote the cyclotomic polynomial of order $p-1$. Show that g is a solution of the congruence $\Phi_{p-1}(x) \equiv 0 \pmod{p}$ if and only if g is a primitive root $\pmod{p}$. Slow also that the sum of all the primitive roots $\pmod{p}$ is $\equiv \mu(p-1) \pmod{p}$.

- p의 primitive root는 $\Phi_{p-1}(x)$의 근과 일치

(i) 차수 $= \phi(p)$

(ii) 최고차항 $= 1$

$\Rightarrow$ (Vieta) $\sum$ primitive root $= -(x^{\phi(p-1)-1}$의 계수$)$

- $x^n - 1 = 0$의 근의 합 $= \begin{cases} 0 & n > 1 \\ 1 & n = 0 \end{cases}$

- $x^n - 1 = \prod_{d|n} \Phi_d(x)$ therefore $S(n) = \sum_{d|n}$ (sum of roots of $\Phi_d(x)$) $\therefore R(d)$

- (möbius) $R(n) = \sum_{d|n} S(d)\, \mu\left(\frac{n}{d}\right)$

$= S(1)\,\mu(n) = \mu(n)$

- sums of primitive roots $= -(-\mu(p-1))$

$\equiv \mu(p-1) \pmod{p}$

9. Let u_0 and u_1 be given, and for $n \geqslant 2$ put $u_n = (u_{n-1} + u_{n-2})/2$. Show that $\lim_{n\to\infty} u_n$ exists, and that it is a certain weighted average of u_0 and u_1.

- $u_n = \dfrac{u_{n-1} + u_{n-2}}{2}$ implies $u_n - u_{n-1} = -\frac{1}{2}(u_{n-1} - u_{n-2})$

- $u_n = u_0 + \sum_{k=1}^{n} (u_1 - u_0)\left(-\frac{1}{2}\right)^{k-1}$

$$\lim_{n\to\infty} u_n = u_0 + (u_1 - u_0) \underbrace{\sum_{k=0}^{\infty} \left(-\frac{1}{2}\right)^k}_{= 1\cdot\left(\frac{1}{1+\frac{1}{2}}\right) = \frac{2}{3}} = \frac{1}{3}u_0 + \frac{2}{3}u_1$$

---

10. If the Euclidean algorithm is applied to the positive integers b and c, $b \geqslant c$, then $r_j = (b, c)$ for some j, and $r_{j+1} = 0$. Put $E(b, c) = j + 1$, so that $E(b, c)$ is the number of divisions performed in executing the algorithm. Show that $E(F_{n+2}, F_{n+1}) = n$ for all positive integers n.

Prove that $r_j \geqslant F_2$, $r_{j-1} \geqslant F_3$, $r_{j-2} \geqslant F_4, \cdots$, and that $b \geqslant F_{j+3}$. More generally, prove that if $F_{n+2} \geqslant b \geqslant c$, then $E(b, c) \leqslant n$, with equality if and only if $b = F_{n+2}$ and $c = F_{n+1}$. Conclude that if $b \geqslant c$ then $E(b, c) < (\log b)/\log((1 + \sqrt{5})/2)$. (This bound was given by Gabriel Lamé in 1845. It was the first occasion in which the worst-case running time of a mathematical algorithm was precisely determined.)

$b = r_{-1} \geq F_{(j+1)+2}$

$\uparrow$

$r_{j-1} \geq F_{1+2}$

$\uparrow$

$r_{j-2} = \delta_j r_{j-1} + r_j$

$\geq 1 \cdot F_3 + F_2 = F_4$

$\exists (\leq c + 3)$

$F_{n+2} = 1 \cdot F_{n+1} + F_n$

$F_{n+1} = 1 \cdot F_{n+2} + F_{n-1}$

$\cdots$

$F_4 = 1 \cdot F_3 + F_2$

$F_3 = 2 \cdot F_2 + 0$

$\# \text{Euclid} = n$

$r_{j-1} \geq F_{1+2}$ and $b \geq F_{j+3}$

$r_{-1} = \delta_1 r_0 + r_1$ $\quad \delta_1 \geq 1$

$r_0 = \delta_2 r_1 + r_2$ $\quad \delta_2 \geq 1$

$\cdots$

$r_{j-2} = \delta_j r_{j-1} + r_j$ $\quad \delta_j \geq 1$

$r_{j-1} = \delta_{j+1} r_j + 0 \Rightarrow \delta_{j+1} \geq 2$, $\quad r_{j-1} = \delta_{j+1} r_j \geq 2 \cdot 1 = F_3$

$(r_{j-1} > r_j > 0)$ $\quad r_j \geq 1 = F_2$

with $E(b, c) = j + 1$

$F_{n+2} \geq b \geq F_{j+3} \Rightarrow F_{n+2} \geq F_{j+3} \Rightarrow E(b, n) \leq n$

가정 $\quad$ 2번 증명

$n \geq j + 1$

if $E(b,c)=n$ then $j+1=n$

$j=n-1$

(그냥 피보나치)

with $F_{n+2} \geq b \geq F_{j+3}$ $\rightsquigarrow$ $\delta_{j+1}=2$ $\rightsquigarrow$ $c=F_{n+1}$

$= F_{n+2}$.

$\delta_{i<j+1}=1$

$b = F_{n+2}$.

$n=E(b,c)$ then $b \geq F_{n+2}$. $> \left(\frac{1+\sqrt{5}}{2}\right)^n$

$\log b > n \log\left(\frac{1+\sqrt{5}}{2}\right) \Rightarrow n < \frac{\log b}{\log\left(\frac{1+\sqrt{5}}{2}\right)}$

$\parallel$

$E(b,c)$

---

11. Extend the method used to prove Theorem 4.10 to derive a formula for u_n if $u_0 = 1$, $u_1 = 2$, $u_2 = 1$, and $u_n = u_{n-1} + 4u_{n-2} - 4u_{n-3}$ for all integers $n \geqslant 3$.

- $u_n - u_{n-1} - 4u_{n-2} + 4u_{n-3} = 0$
- suppose $u_n = r^n$ $\quad r^3 - r^2 - 4r + 4 = 0$

  $(r-2)(r+2)(r-1) = 0$
- 일반해 $u_n = A + B(2^n) + C(-2)^n$

$u_0 = 1$

$u_1 = 2$ implies $u_n = 1 + \frac{1}{4}(2)^n - \frac{1}{4}(-2)^n$

$u_2 = 1$

***20.** Show that if p is an odd prime then $U_p \equiv \left(\frac{D}{p}\right) \pmod{p}$.

$x^2 - Px + Q = 0$

$\Rightarrow$ root $\alpha = \frac{P+\sqrt{D}}{2}$ $\beta = \frac{P-\sqrt{D}}{2}$ $\Rightarrow U_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}$

$$\Rightarrow U_p = \frac{1}{\sqrt{D}}\left[\left(\frac{P+\sqrt{D}}{2}\right)^p - \left(\frac{P-\sqrt{D}}{2}\right)^p\right]$$

$$= \frac{1}{2^p\sqrt{D}}\left(\sum_{k=0}^{p}\binom{p}{k} P^{p-k}\left((\sqrt{D})^k + (-\sqrt{D})^k\right)\right)$$

$$= \frac{1}{2^{p-1}}\left[\binom{p}{1}P^{p-1} + \binom{p}{3}P^{p-3}D^1 + \cdots + \binom{p}{p}D^{\frac{p-1}{2}}\right]$$

$\Rightarrow$ if $1 < k < p$ then $\binom{p}{k} \equiv 0 \pmod{p}$

therefore $U_p \equiv \underbrace{\frac{1}{2^{p-1}}}_{\substack{\shortparallel \\ 1}} \cdot \underbrace{\binom{p}{k}}_{\substack{\shortparallel \\ 1}} \cdot \underbrace{D^{\frac{p-1}{2}}}_{\substack{\shortparallel \\ \left(\frac{D}{p}\right)}} \pmod{p}$

Fermat 이항정리 Euler Criteria

***27.** Let the sequence u_n be determined by the relations $u_1 = 0$, $u_2 = 2$, $u_3 = 3$, and $u_{n+1} = u_{n-1} + u_{n-2}$ for $n \geqslant 3$. Prove that if p is prime then $p | u_p$. (The least composite number with this property is 271,441 $= 521^2$.)

- $u_n = u_{n-2} + u_{n-3}$

- $x^3 - x - 1 = 0 \rightsquigarrow \alpha + \beta + \gamma = 0$

  $\alpha\beta + \beta\gamma + \gamma\alpha = -1$

  $\alpha\beta\gamma = 1$

- $u_1 = 0$, $u_2 = 2$, $u_3 = 3$ implies $S_n = \alpha^n + \beta^n + \gamma^n$

- $(\alpha+\beta+\gamma)^p = \alpha^p + \beta^p + \gamma^p + p \cdot Q(\alpha, \beta, \gamma)$

  $0 = u_p + p \cdot Q(\alpha, \beta, \gamma)$ Hence $u_p \equiv 0 \pmod p$

---

1. Given any m integers none of which is a multiple of m, prove that two can be selected whose difference is a multiple of m.

$\Longleftrightarrow$ $a_1, \ldots, a_m$ s.t. $\forall a_i \not\equiv 0 \pmod m$

then $\exists k, l$ s.t. $a_k \equiv a_l \pmod m$

otherwise $a_1 \in \{1, \ldots, m-1\}$

$a_2 \in \{1, \ldots, m-1\} \setminus \{a_1\}$

$\ldots$

$a_m \in \{1, \ldots, m\} \setminus \{a_1, \ldots, a_{m-1}\}$

$= \emptyset$ $\rightarrow\leftarrow$

***2.** If $\mathscr{S}$ is any set of $n+1$ integers selected from $1, 2, 3, \cdots, 2n+1$, prove that $\mathscr{S}$ contains two relatively prime integers. Prove that the result does not hold if $\mathscr{S}$ contains only n integers.

- suppose 1 not included (trivial)

  $\{2,3\}$ $\{4,5\}$ $\{6,7\}$ $\cdots$ $\{2n, 2n+1\}$ (n개 pair)

  1개의 pair는 중복해서 들어감 $\Rightarrow$ 어떤도 중복!

- $\{2, 4, \ldots, 2n\}$

***3.** For any positive integers k and $m > 1$, let $\mathscr{S}$ be a set of k integers none of which is a multiple of m. If $k > m/2$, prove that there are two integers in $\mathscr{S}$ whose sum or whose difference is divisible by m.

m 홀수 $\{1, \ldots, r, -r, \ldots, -1\}$

$\Downarrow$

$k \geq r+1$

$|a_1| \in \{1, \ldots, r\}$

$|a_2| \in \{1, \ldots, r\} \setminus \{|a_1|\}$

$\Downarrow$

$|a_{r+1}| \in \{1, \ldots, r\} \setminus \{|a_1| \cdots |a_r|\}$

$= \emptyset$ $\rightarrow\leftarrow$,

m 짝수 $\{1, \ldots, r, r+1, -r, \ldots, -1\}$.

$\Downarrow$

$k \geq r+2$.

$|a_1| \in \{1, \ldots, r+1\}$

$|a_2| \in \{1, \ldots, r+1\} \setminus \{|a_1|\}$

$\Downarrow$

$|a_{r+1}| \in \{1, \ldots, r+1\} \setminus \{|a_1| \cdots |a_{r+1}|\}$

$= \emptyset$ $\rightarrow\leftarrow$,

***4.** Let the integers $1, 2, \cdots, n$ be placed in any order around the circumference of a circle. For any $k < n$, prove that there are k integers in a consecutive block on the circumference having sum at least $(kn + k)/2$.

seq $a_1 \ldots a_n$ $\Rightarrow$ consider

$\underbrace{(a_1 + \cdots + a_k)}_{b_1}$

$\underbrace{+ (a_2 + \cdots + a_{k+1})}_{b_2}$

$\cdots \underbrace{(a_n + a_1 + \cdots + a_{k-1})}_{b_n}$

$= k(a_1 + \cdots + a_{k+1})$

$\Rightarrow$ average (k개 합)

$= \frac{1}{n} \cdot k\left(\frac{n(n+1)}{2}\right)$

$= \frac{k(n+1)}{2}$.

i.e $\mathbb{E}[b] = \frac{k(n+1)}{2}$.

if $\forall b_i < \frac{k(n+1)}{2}$

then $\mathbb{E}[b] < \frac{k(n+1)}{2}$ $\rightarrow\leftarrow$

***5.** Given any integers a, b, c and any prime p not a divisor of ab, prove that $ax^2 + by^2 \equiv c \pmod{p}$ is solvable.

$p = 2$ then $ax^2 + by^2 \equiv 0$ if $x \equiv 0$ $y \equiv 0$

$\equiv 1$ if $x \equiv 0$ $y \equiv 1$ $\because$ b is odd.

- $x^2, y^2 \equiv 0, \underset{\substack{\uparrow \\ \text{primitive root}}}{l^2}, l^4 \ldots l^{2r} \pmod{p}$ with $r = \frac{p-1}{2}$.

- $ax^2 \equiv c - by^2$ since $(r+1) + (r+1) > p = 2r+1$

$\Downarrow$ $r+1$개 $\Downarrow$ $r+1$개.

비둘기집 $\Rightarrow \{ax^2\} \cap \{c - by^2\} \neq \emptyset$